WORKSHEET 6.2

Riemann Sums and the Trapezoidal Rule

Unit 6: Integration and Accumulation of Change

Key ideas

- A **left**, **right** or **midpoint** Riemann sum adds rectangles whose heights are taken at the left end, right end or midpoint of each subinterval.
- A **trapezoidal sum** uses the average of the two end heights on each subinterval.
- With a table, use the table's own subintervals. The widths are often **unequal**.
- If f is increasing, a left sum underestimates and a right sum overestimates; if f is decreasing, the reverse.

Worked examples

Example 1 · Equal widths

Approximate $\displaystyle\int_0^3 \left(x^2+1\right)\,dx$ with left and right Riemann sums using 3 subintervals.

Solution. Width 1. Left: $f(0)+f(1)+f(2)=1+2+5$. Right: $f(1)+f(2)+f(3)=2+5+10$.

Left sum 8; right sum 17.

Example 2 · Unequal widths

Selected values of h are shown.

t	0	1	4	6
$h(t)$	2	5	9	7

Approximate $\displaystyle\int_0^6 h(t)\,dt$ with a trapezoidal sum.

Solution. $\frac{1}{2}(2+5)(1)+\frac{1}{2}(5+9)(3)+\frac{1}{2}(9+7)(2)=3.5+21+16$.

40.5**.**

Example 3 · Over or under

A function is increasing on $[a,b]$. Compare its left and right Riemann sums with the integral.

Solution. Each left rectangle lies below the graph and each right rectangle above it.

Left sum $<$ integral $<$ right sum.

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Practice

Questions 5 to 7 use this table of values of a continuous function R.

t	0	2	3	7	10
$R(t)$	6	10	12	8	2

CORE

1. Find the left Riemann sum for $f(x) = 2x$ on $[0, 4]$ with 4 equal subintervals.

2. Find the right Riemann sum for $f(x) = 2x$ on $[0, 4]$ with 4 equal subintervals.

3. Find the midpoint Riemann sum for $f(x) = x^2$ on $[0, 4]$ with 2 equal subintervals.

4. A function is decreasing on $[a, b]$. Does a left Riemann sum overestimate or underestimate the integral?

EXAM LEVEL

5. Approximate $\int_0^{10} R(t)\,dt$ with a left Riemann sum using the four subintervals of the table.

6. Approximate the same integral with a right Riemann sum.

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7. Approximate the same integral with a trapezoidal sum.

8. Find the midpoint Riemann sum for $f(x) = \dfrac{1}{x}$ on $[1, 5]$ with 2 equal subintervals.

Question 9 · Free-response practice

The speed of a train, $v(t)$ metres per second, is recorded at times t seconds.

t	0	10	25	40	60
$v(t)$	0	12	20	16	4

(a) Approximate $\displaystyle\int_0^{60} v(t)\,dt$ with a trapezoidal sum using the four subintervals of the table.

(b) Using correct units, explain the meaning of the integral in part (a).

(c) Use your answer to estimate the average speed of the train over the 60 seconds.

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Worked solutions

1. $1(0+2+4+6) = 12$.
2. $1(2+4+6+8) = 20$.
3. Midpoints 1 and 3, width 2: $2(1+9) = 20$.
4. It overestimates.
5. $(2)(6) + (1)(10) + (4)(12) + (3)(8) = 94$.
6. $(2)(10) + (1)(12) + (4)(8) + (3)(2) = 70$.
7. $\frac{1}{2}(6+10)(2) + \frac{1}{2}(10+12)(1) + \frac{1}{2}(12+8)(4) + \frac{1}{2}(8+2)(3) = 16+11+40+15 = 82$.
8. Midpoints 2 and 4, width 2: $2\left(\frac{1}{2}+\frac{1}{4}\right) = 1.5$.
9. (a) $\frac{1}{2}(0+12)(10) + \frac{1}{2}(12+20)(15) + \frac{1}{2}(20+16)(15) + \frac{1}{2}(16+4)(20) = 60+240+270+200 = 770$.
   (b) The distance the train travels, in metres, during the 60 seconds.
   (c) $\frac{770}{60} \approx 12.8$ metres per second.

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